O*NET OnLine

Database Administrators

15-1242.00

Updated 2026

Administer, test, and implement computer databases, applying knowledge of database management systems. Coordinate changes to computer databases. Identify, investigate, and resolve database performance issues, database capacity, and database scalability. May plan, coordinate, and implement security measures to safeguard computer databases.

Sample of reported job titles: Database Administration Manager, Database Administrator (DBA), Database Analyst, Database Coordinator, Database Engineer, Database Manager, Development and Database Administration Manager, IS Admin (Information Systems Administrator), Systems Administrator (Systems Admin)

Summary | Details | Custom | Easy Read | Veterans | Español

Contents

Occupation-Specific Information

Tasks

Save Table: XLSX CSV

10 of 18 displayed

Importance	Category	Task
75	Core	Modify existing databases and database management systems or direct programmers and analysts to make changes.
73	Core	Plan, coordinate, and implement security measures to safeguard information in computer files against accidental or unauthorized damage, modification or disclosure.
72	Core	Plan and install upgrades of database management system software to enhance database performance.
70	Core	Specify users and user access levels for each segment of database.

Importance	Category	Task
69	Core	Test changes to database applications or systems.
67	Core	Test programs or databases, correct errors, and make necessary modifications.
64	Core	Train users and answer questions.
62	Core	Provide technical support to junior staff or clients.
61	Core	Approve, schedule, plan, and supervise the installation and testing of new products and improvements to computer systems, such as the installation of new databases.
61	Core	Develop standards and guidelines for the use and acquisition of software and to protect vulnerable information.

Occupational Requirements

Work Activities

Save Table: XLSX CSV

10 of 41 displayed

Importance	Work Activity
96	**Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
80	**Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.
78	**Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
78	**Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
75	**Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.

Importance	Work Activity
73	**Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
72	**Establishing and Maintaining Interpersonal Relationships** — Developing constructive and cooperative working relationships with others, and maintaining them over time.
71	**Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.
69	**Evaluating Information to Determine Compliance with Standards** — Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards.
68	**Thinking Creatively** — Developing, designing, or creating new applications, ideas, relationships, systems, or products, including artistic contributions.

Detailed Work Activities

Save Table: XLSX CSV

10 of 21 displayed

- Create databases to store electronic data.
- Update computer database information.
- Implement security measures for computer or information systems.
- Develop computer or information security policies or procedures.
- Install computer software.
- Assess database performance.
- Test computer system operations to ensure proper functioning.
- Modify software programs to improve performance.
- Train others in computer interface or software use.
- Provide technical support for software maintenance or use.

Work Context

Save Table: XLSX CSV

10 of 57 displayed

- **E-Mail** — How frequently does your job require you to use E-mail?

92% Every day

- **Indoors, Environmentally Controlled** — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?

  90% Every day

- **Telephone Conversations** — How often do you have telephone conversations in this job?
- **Work With or Contribute to a Work Group or Team** — How important is it to work with or contribute to a work group or team in this job?
- **Freedom to Make Decisions** — How much decision making freedom, without supervision, does the job offer?

  56% A lot of freedom

  38% Some freedom

- **Importance of Being Exact or Accurate** — How important is being very exact or highly accurate in performing this job?
- **Spend Time Sitting** — How much does this job require sitting?

  24% More than half the time

- **Determine Tasks, Priorities and Goals** — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?
- **Contact With Others** — How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?

  23% Constant contact with others

  71% Contact with others most of the time

- **Frequency of Decision Making** — How often is the worker required to make decisions that affect other people, the financial resources, and/or the image and reputation of the organization?

  17% Every day

  70% Once a week or more but not every day

Experience Requirements

Job Zone

Save Table: XLSX CSV

Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, conservation scientists, art directors, and cost estimators.
SVP Range	(7.0 to < 8.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

Apprenticeship Opportunities

All 2 displayed

Example apprenticeship titles for this occupation:

- Database Administrator (Nof)
- Database Technician

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

Save Table: XLSX CSV

10 of 63 displayed

- **Backup or archival software** — Acronis Recovery Expert; EMC NetWorker; HP DataProtector; Veritas NetBackup
- **Business intelligence and data analysis software** — Apache Spark ; IBM Cognos Impromptu; Microsoft Power BI ; Tableau
- **Data base management system software** — Amazon DynamoDB ; Elasticsearch ; MongoDB ; Oracle PL/SQL
- **Data base user interface and query software** — Apache Hive ; Blackboard software; IBM DB2 ; MySQL
- **Development environment software** — Apache Kafka ; Apache Maven ; Go ; Oracle SQL Developer
- **Enterprise resource planning ERP software** — Ellucian Banner ERP; Microsoft Dynamics; Oracle PeopleSoft ; SAP software
- **Metadata management software** — IBM Rational Data Architect; Informatica software ; Pentaho Kettle; Quest Erwin Data Modeler
- **Object or component oriented development software** — C# ; jQuery ; Scala ; Swift
- **Operating system software** — Bash ; Microsoft Windows Server ; Red Hat Enterprise Linux ; UNIX Shell
- **Web platform development software** — Django ; Google Angular ; Microsoft ASP.NET ; Spring Framework

Hot Technologies are requirements most frequently included across all employer job postings.
See all 112 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 38 In Demand skills for this occupation.

Essential Skills

Save Table: XLSX CSV

All 10 displayed

Importance	Essential Skill
72	**Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
69	**Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
69	**Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.

Importance	Essential Skill
63	**Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
60	**Speaking** — Talking to others to convey information effectively.
56	**Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
56	**Writing** — Communicating effectively in writing as appropriate for the needs of the audience.
50	**Learning Strategies** — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
50	**Mathematics** — Using mathematics to solve problems.
25	**Science** — Using scientific rules and methods to solve problems.

Transferable Skills

Save Table: XLSX CSV

10 of 25 displayed

Importance	Transferable Skill
72	**Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
69	**Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
60	**Programming** — Writing computer programs for various purposes.
53	**Operations Analysis** — Analyzing needs and product requirements to create a design.
53	**Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
53	**Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.

Importance	Transferable Skill
50	**Coordination** — Adjusting actions in relation to others' actions.
50	**Instructing** — Teaching others how to do something.
47	**Service Orientation** — Actively looking for ways to help people.
47	**Social Perceptiveness** — Being aware of others' reactions and understanding why they react as they do.

Knowledge

Save Table: XLSX CSV

10 of 33 displayed

Importance	Knowledge
95	**Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
74	**English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
73	**Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
64	**Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
64	**Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.
63	**Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services.
53	**Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources

Importance	Knowledge
	modeling, leadership technique, production methods, and coordination of people and resources.
52	**Education and Training** — Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects.
45	**Design** — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.
43	**Economics and Accounting** — Knowledge of economic and accounting principles and practices, the financial markets, banking, and the analysis and reporting of financial data.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

89%	Bachelor's degree required
4%	Post-baccalaureate certificate required
3%	Associate's degree or other 2-year degree required

Worker Characteristics

Abilities

Save Table: XLSX CSV

10 of 52 displayed

Importance	Ability
75	**Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
72	**Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
72	**Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).

Importance	Ability
72	**Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.
72	**Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
72	**Written Comprehension** — The ability to read and understand information and ideas presented in writing.
66	**Near Vision** — The ability to see details at close range (within a few feet of the observer).
63	**Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
60	**Category Flexibility** — The ability to generate or use different sets of rules for combining or grouping things in different ways.
60	**Written Expression** — The ability to communicate information and ideas in writing so others will understand.

Career Interest Types

Save Table: XLSX CSV

All 6 displayed

Occupational Interest	Career Interest Type
94	**Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
64	**Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
32	**Enterprising** — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal

Occupational Interest	Career Interest Type
	situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law.
31	**Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
22	**Social** — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities.
3	**Artistic** — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art.

Specific Interest Areas

Save Table: XLSX CSV

3 of 41 displayed

Occupational Interest	Specific Interest Area
95	**Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
44	**Management/Administration** — Work involves planning, directing, or coordinating the daily operations of an organization, and overseeing personnel, business, or administrative services.
37	**Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.

Work Styles

Save Table: XLSX CSV

10 of 20 displayed

Impact	Work Style
100	**Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
88	**Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.
82	**Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
77	**Integrity** — A tendency to be honest and ethical at work.
73	**Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
59	**Achievement Orientation** — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards.
57	**Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work.
56	**Perseverance** — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work.
49	**Initiative** — A tendency to be proactive and take on extra responsibilities and tasks that may fall outside of one's required work role.
44	**Tolerance for Ambiguity** — A tendency to be comfortable with ambiguity and uncertainty at work.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025) $50.30 hourly, $104,620 annual

State wages Select a State Go

Local wages ZIP Code Go

Employment (2025)	75,000 employees
Projected growth (2025-2035)	Little or no change
Projected job openings (2025-2035)	3,300
State trends	Select a State Go
Top industries (2025)	Professional, Scientific, and Technical Services (23% employed in this sector) Finance and Insurance (12%) Educational Services (11%)

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings	Select a State Go
Local job openings	ZIP Code Go

More Information

Related Occupations

Save Table: XLSX CSV

10 of 20 displayed

- 15-1211.00 Computer Systems Analysts Bright Outlook
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1243.01 Data Warehousing Specialists
- 15-1243.00 Database Architects
- 15-1299.03 Document Management Specialists
- 15-1212.00 Information Security Analysts
- 15-1299.05 Information Security Engineers
- 15-1244.00 Network and Computer Systems Administrators
- 15-1252.00 Software Developers
- 15-1299.01 Web Administrators

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

- Association for Computing Machinery
- Computing Research Association
- IEEE Computer Society
- National Center for Women and Information Technology

Accreditation, Certification, & Unions

- CompTIA
- Institute for Certification of Computing Professionals